

CARBON CYCLE

Major role of particle fragmentation in regulating biological sequestration of CO₂ by the oceans

Nathan Briggs^{1,2*}, Giorgio Dall'Olmo^{3,4}, Hervé Claustre²

A critical driver of the ocean carbon cycle is the downward flux of sinking organic particles, which acts to lower the atmospheric carbon dioxide concentration. This downward flux is reduced by more than 70% in the mesopelagic zone (100 to 1000 meters of depth), but this loss cannot be fully accounted for by current measurements. For decades, it has been hypothesized that the missing loss could be explained by the fragmentation of large aggregates into small particles, although data to test this hypothesis have been lacking. In this work, using robotic observations, we quantified total mesopelagic fragmentation during 34 high-flux events across multiple ocean regions and found that fragmentation accounted for $49 \pm 22\%$ of the observed flux loss. Therefore, fragmentation may be the primary process controlling the sequestration of sinking organic carbon.

Large organic particles (>100 μm) sinking through the ocean's mesopelagic zone (100 to 1000 m) play a critical role in regulating the global carbon cycle. These particles are part of the biological carbon pump, which transfers an estimated 5 to 12 Pg of C per year (1–3) from the sunlit ocean and sequesters 15 to 30% of this carbon for centuries to millennia in the deep ocean (4–6). The organic carbon that is sequestered directly affects atmospheric CO₂ concentrations (7). Sinking organic carbon is also a primary source of energy for ocean ecosystems in and below the mesopelagic zone and is essential to the ecosystem services they provide (8, 9).

Despite its importance, we still lack a quantitative, mechanistic understanding of key parts of the biological carbon pump. Particularly, we poorly understand the subsurface loss processes that determine the depth at which sinking organic carbon is remineralized to CO₂. This depth affects, in turn, long-term atmospheric CO₂ sequestration (7). Measurements of sinking particle flux at different depths via underwater sediment traps, radioactive particle tracers, and underwater cameras indicate that, on average, ~70 to 85% of sinking carbon flux is lost in the mesopelagic (4–6). However, direct consumption of fast-sinking particles, either by attached bacteria (10–12) or by zooplankton (13), appears to explain less than half of this observed flux attenuation. The remaining $\geq 50\%$ of the observed mesopelagic flux attenuation might be explained by fragmentation into smaller, slower-sinking particles (12). This would be consistent with the observed seasonal buildup of small particles in the mesopelagic zone (14) and the metabolic

activities of free-living bacteria that consume such particles (13). Fragmentation rates have been estimated more directly, in both the laboratory and the upper ocean, from changes in particle size and have been attributed to several mechanisms. Microbial degradation has been shown to fragment marine particles in the laboratory (15, 16), marine particle fragmentation by zooplankton feeding has been observed both in the laboratory (17) and in the upper ocean (18), and fragmentation caused by ocean turbulence has been proposed to explain patterns of particle size in the mixed layer (19, 20). In the mesopelagic zone, however, similar studies have not been practical.

As a consequence, the hypothesis that fragmentation can reconcile existing measurements of the mesopelagic carbon budget has not been rigorously tested. To address this, we have estimated fragmentation rates at broad scale by simultaneously tracking changes in large (21) and small (14) mesopelagic particle concentrations using optical data collected by Biogeochemical-Argo floats.

We analyzed data from 25 floats deployed across the subpolar North Atlantic and the Atlantic and Indian sectors of the Southern Ocean between 2013 and 2016 (Fig. 1). All floats carried sensors for particulate optical backscattering (b_{bp}), a proxy for particulate mass concentration (22), and for chlorophyll *a* fluorescence (F), a proxy for live phytoplankton biomass (23) (see table S1 for full list of abbreviations used in this manuscript). Floats were profiled to 1000 m with temporal and vertical resolutions of 2 to 3 days and 1 m, respectively, during spring and summer phytoplankton blooms. F and b_{bp} were each divided into three components (fig. S1) [as described in (24)]: deep sensor blanks, including a background of small refractory particles (b_{br} and F_{r}); small, labile backscattering (b_{bs}) and fluorescing (F_{s}) particles; and large, fast-sinking backscattering (b_{bl}) and fluorescing (F_{l}) particles. The division between small and large corresponds roughly to a particle diameter of 100 μm for b_{bs} versus b_{bl} and a particle chlorophyll content of 60 pg for F_{s} versus F_{l} (24). We attribute F_{l} primarily to live phytoplankton

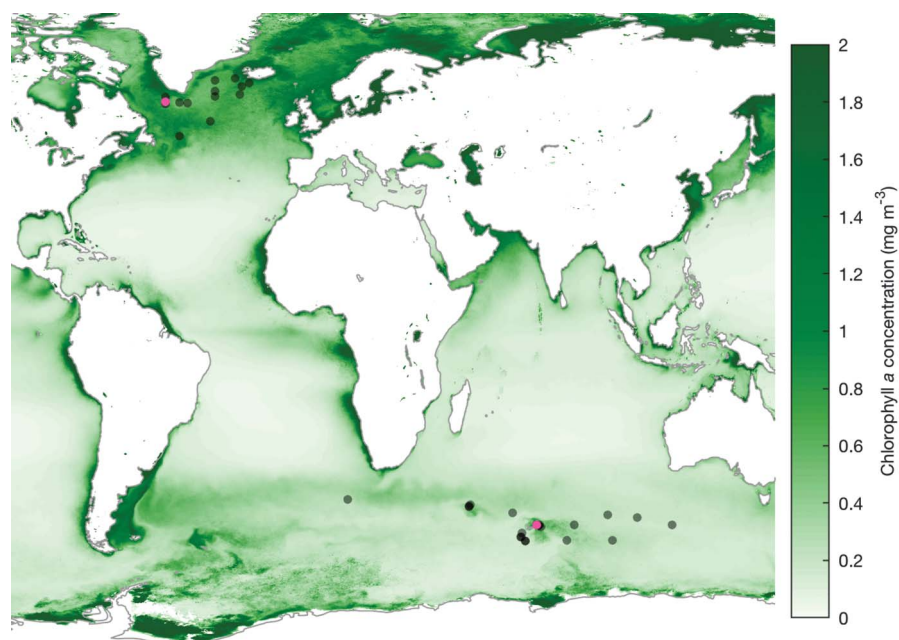


Fig. 1. Location of particle-flux pulses. Gray circles represent the 34 pulses analyzed in this study. Darker grays indicate overlapping circles. Magenta circles indicate example pulses shown in Fig. 2. The background is a map of climatological mean surface chlorophyll concentration from MODIS-Aqua (2002 to 2017).

¹National Oceanography Centre, Southampton, UK.

²Laboratoire d'Océanographie de Villefranche-sur-mer (LOV), Sorbonne Université and CNRS Villefranche-sur-Mer, France.

³Plymouth Marine Laboratory, Plymouth, UK. ⁴National Centre for Earth Observations, Plymouth, UK.

*Corresponding author. Email: natebriggs@gmail.com

aggregates; F_l represents a subset of b_{bl} , which additionally includes fecal and detrital matter (21). Between May 2013 and February 2018, we identified 34 pulses of b_{bl} and/or F_l in the mesopelagic that were associated with surface phytoplankton blooms and were clearly distinguishable from prebloom background concentrations. Bulk large-particle sinking velocity was estimated for each large-particle pulse (fig. S2) from the timing of peak concentration versus depth (24). Mean sinking velocities (and 95% confidence intervals) across all pulses were 74 (58 to 100) m per day for large backscattering particles and 98 (79 to 129) m per day for large fluorescing particles.

We observed close coupling between large- and small-particle concentrations during these flux pulses (Fig. 2). Small-particle concentrations increased rapidly during periods of peak large-particle concentration (Fig. 2; solid black lines) at all depths below 200 m, peaking slightly later (e.g., Fig. 2, left column: peak F_s lags behind

peak F_l by ~2 days, regardless of depth). This coupling provides strong evidence that large-particle fragmentation drives the observed accumulation of small particles in the mesopelagic, both for large particles in general (b_{bl}) and phytoplankton aggregates in particular (F_l).

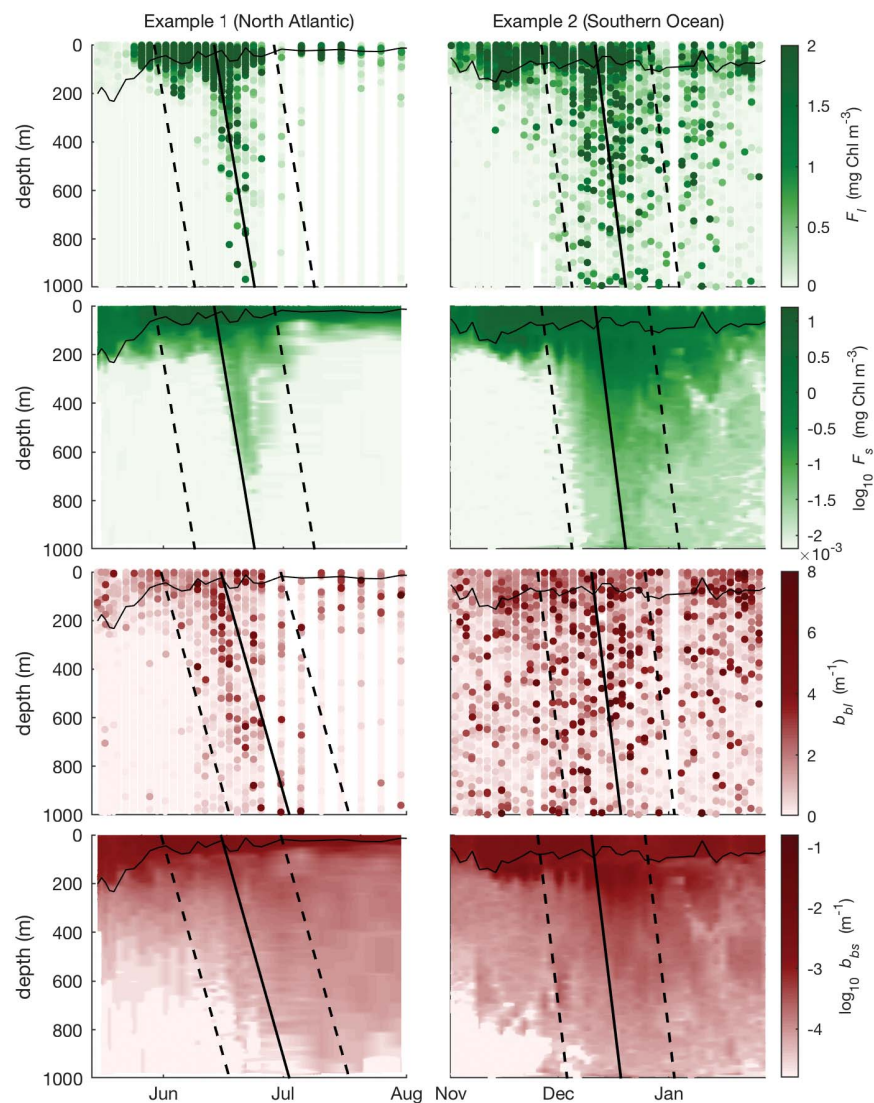
We quantified specific fragmentation rates during each sinking pulse by tracking these changes in the concentrations of large and small particles as a function of depth and time. Full computations, assumptions, and uncertainty budgets (24) are shown in figs. S3 to S11 along with alternative calculations supporting key methodological assumptions (figs. S11 to S13). Mean fragmentation rate profiles across all pulses varied with depth and particle type from 0.03 to 0.27 per day (Fig. 3). Although wide uncertainty bounds prevent firm conclusions, the patterns in these rates offer preliminary indications of possible fragmentation mechanisms. First, live phytoplankton aggre-

gates (F_l) fragmented at higher rates than large sinking particles in general (b_{bl}) at all depths in the mesopelagic zone (Fig. 3). Fresh phytoplankton aggregates therefore appear either more fragile than other large sinking particles and/or are subject to higher local shear. The latter might result from selective feeding on fresh material by zooplankton. Second, specific fragmentation rates decreased with depth (Fig. 3). This depth dependency could result from passive breakup of more fragile particles closer to the surface. It might also result from higher zooplankton activity closer to the surface, where we expect food to be more abundant and more nutritious. On average, fragmentation accounted for close to 50% of the observed loss rates of large particles in general and 30 to 60% of the loss of large fluorescing particles (Fig. 3) at all depths between 250 and 950 m.

We also found regional differences in specific fragmentation rates. When calculated

Fig. 2. Fragmentation of large particles generates small particles at depth. Large- and small-particle measurements from example large-particle pulses from the North Atlantic (left panels) and the Southern Ocean (right panels) are shown.

Large-particle fluorescence F_l (green circles) and large-particle backscattering b_{bl} (red circles) are shown above the corresponding $\log_{10}$ small-particle fluorescence F_s (green) and backscattering b_{bs} (red). Large-particle measurements are plotted individually with higher values (darker colors) covering lower values. Thin black lines along the top edges of the panels show mixed-layer depth; thick, diagonal solid lines show linear least-squares fits of maximum large-particle concentration with depth; and dashed lines show the ± 15 day windows used for fragmentation calculations. Similar visualizations for all 34 plumes in this study can be found at seano.e.org (26). Chl, chlorophyll.



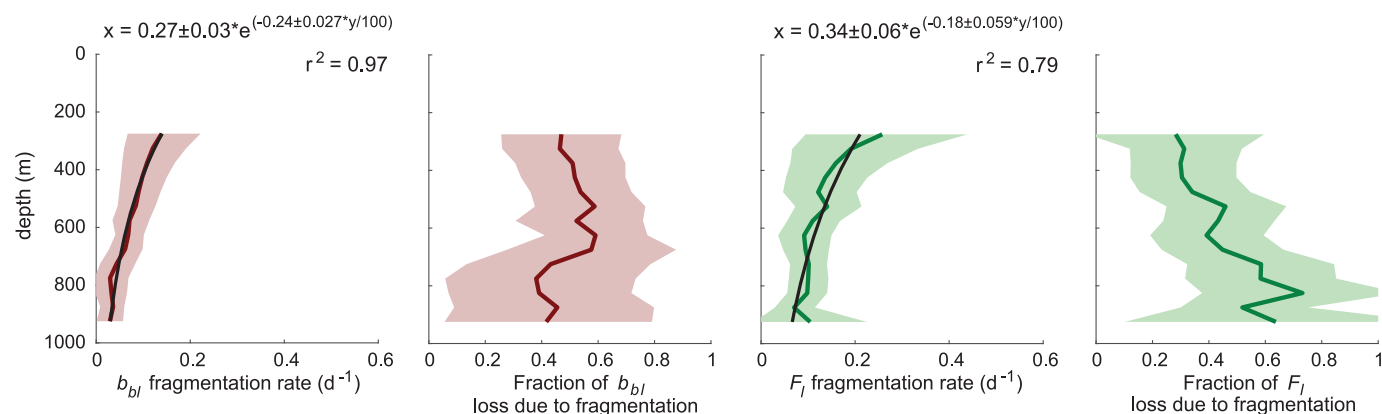


Fig. 3. Fragmentation contributes 50% of the observed flux attenuation. First and third panels from left show mean specific fragmentation rates of large particles b_{bl} (red) and of large fluorescing particles F_l (green) across all large-particle pulses. Second and fourth panels from left show the mean fraction of b_{bl} flux attenuation (red) and

F_l flux attenuation (green) explained by this fragmentation. Shaded areas show 95% confidence intervals. Black curves and equations in first and third panels show least-squared exponential fits of specific fragmentation rates versus depth. d, day; r^2 , coefficient of determination; x, specific fragmentation rate (per day); y, depth (m).

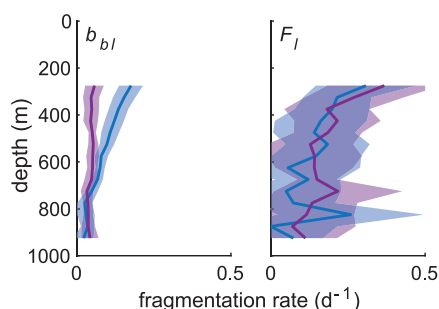


Fig. 4. Regional differences in fragmentation.

Comparison between mean specific fragmentation rates of large particles b_{bl} (left) and those of large fluorescing particles F_l (right) during North Atlantic (purple) and Southern Ocean (blue) phytoplankton blooms. Bold lines show means and shaded areas show two standard errors around the means.

using the same parameterizations (24), large-particle (b_{bl}) specific fragmentation rates were notably higher in the Southern Ocean than those in the North Atlantic, between 250 and 600 m (Fig. 4, left panel). On the other hand, fragmentation of fresh phytoplankton aggregates (F_l) was not different in the two regions (Fig. 4, right panel). Further differences in b_{bl} fragmentation were observed between subregions of the Southern Ocean (table S2). Investigation of these regional differences may help to constrain the drivers of fragmentation.

Our measurements provide quantitative and geographically broad support for the hypothesis that fragmentation exerts a major control on mesopelagic carbon flux (12), which has two notable implications. First, when added to previous estimates of large-particle consumption by zooplankton and bacteria (13), fragmentation can now fully explain the observed

flux attenuation at high latitudes. Therefore, these results strengthen our mechanistic understanding of the biological carbon pump. Second, our results imply that fragmentation may be the single most important process in determining the depth at which fast-sinking organic carbon is remineralized. By extension, fragmentation appears to be a key regulator of atmospheric CO_2 concentrations (7) and of the delivery of energy to deep-ocean ecosystems versus its retention in mesopelagic ecosystems (25).

REFERENCES AND NOTES

1. P. W. Boyd, T. W. Trull, *Prog. Oceanogr.* **72**, 276–312 (2007).
2. S. A. Henson et al., *Geophys. Res. Lett.* **38**, L04606 (2011).
3. D. A. Siegel et al., *Global Biogeochem. Cycles* **28**, 181–196 (2014).
4. J. H. Martin, G. A. Knauer, D. M. Karl, W. W. Broenkow, *Deep Sea Res. Part A Oceanogr. Res. Pap.* **34**, 267–285 (1987).
5. L. Guidi et al., *Global Biogeochem. Cycles* **29**, 1044–1059 (2015).
6. S. A. Henson, R. Sanders, E. Madsen, *Global Biogeochem. Cycles* **26**, GB1028 (2012).
7. E. Y. Kwon, F. Primeau, J. L. Sarmiento, *Nat. Geosci.* **2**, 630–635 (2009).
8. X. Irigoien et al., *Nat. Commun.* **5**, 3271 (2014).
9. M. A. St. John et al., *Front. Mar. Sci.* **3**, 31 (2016).
10. J. R. Collins et al., *Global Biogeochem. Cycles* **29**, 1471–1494 (2015).
11. A. Belcher et al., *Biogeosciences* **13**, 4927–4943 (2016).
12. D. M. Karl, G. A. Knauer, J. H. Martin, *Nature* **332**, 438–441 (1988).
13. S. L. C. Giering et al., *Nature* **507**, 480–483 (2014).
14. G. Dall'Olmo, K. A. Mork, *Geophys. Res. Lett.* **41**, 2921–2927 (2014).
15. L. R. Pomeroy, D. Deibel, *Limnol. Oceanogr.* **25**, 643–652 (1980).
16. B. A. Biddanda, L. R. Pomeroy, *Mar. Ecol. Prog. Ser.* **42**, 79–88 (1988).
17. S. Goldthwait, J. Yen, J. Brown, A. Alldredge, *Limnol. Oceanogr.* **49**, 940–952 (2004).
18. L. Dilling, A. L. Alldredge, *Deep Sea Res. Part I Oceanogr. Res. Pap.* **47**, 1227–1245 (2000).
19. G. A. Jackson, *Deep Sea Res. Part II Topical Stud. Oceanogr.* **42**, 159–184 (1995).
20. J. Ruiz, *Deep Sea Res. Part I Oceanogr. Res. Pap.* **44**, 1105–1126 (1997).
21. N. Briggs et al., *Deep Sea Res. Part I Oceanogr. Res. Pap.* **58**, 1031–1039 (2011).
22. R. A. Reynolds, D. Stramski, G. Neukermans, *Limnol. Oceanogr.* **61**, 1869–1890 (2016).
23. C. Roessler et al., *Limnol. Oceanogr. Methods* **15**, 572–585 (2017).

24. See supplementary materials.

25. D. J. Mayor, R. Sanders, S. L. C. Giering, T. R. Anderson, *BioEssays* **36**, 1132–1137 (2014).

26. N. Briggs, G. Dall'Olmo, H. Claustre, Size-fractionated optical backscattering and chlorophyll fluorescence from 34 high-latitude phytoplankton blooms. SEANOE (2019); <https://doi.org/10.17882/70484>.

ACKNOWLEDGMENTS

We thank the entire international Argo community for its work over decades to create a reliable global profiling float network and its recent work to integrate biogeochemical sensors into this network. We especially thank A. Poteau for his development and continuous support of float control and visualization tools used to adapt float vertical and temporal sampling resolutions to the present study requirements. We also thank two anonymous reviewers for their thorough and insightful comments, which led to substantial improvements to this manuscript. **Funding:** The collection of the data used in this manuscript was funded by a European Research Council Advanced grant (remOcean, agreement no. 246577) as well as the climate initiative of the BNP Paribas foundation (SOCLIM project), French LEFE-GMMC, and UK BioArgo projects. Analysis was funded by U.S. National Science Foundation Postdoctoral Research Fellowship OCE1420929. Final writing was funded by a European Research Council Consolidator grant (GOCART, agreement no. 724416) and a European Research Council Advanced grant (REFINE, agreement no. 834177). **Author contributions:** N.B. and G.D. conceptualized the study. N.B. developed the methods, and H.C. managed the data collection to optimize for these methods. N.B. carried out all data analysis, including software development, with periodic feedback from G.D. and H.C. N.B. wrote the original draft and generated all figures. N.B., G.D., and H.C. reviewed and edited the final manuscript.

Competing interests: The authors declare no competing interests.

Data and materials availability: All data used in this study are available from the Argo Global Data Assembly Centers in Brest, France (<http://ftp.ifremer.fr/ifremer/argo/dac/coriolis>) and Monterey, California (<http://usgodae.org/pub/outgoing/argo/dac/coriolis>), in subfolders with names corresponding to the WMO numbers of individual floats given in table S2 of the supplemental methods. Intermediate (binned) data products used in this study are available at seanoe.org (26), along with data processing visualizations for all 34 plumes.

SUPPLEMENTARY MATERIALS

science.sciencemag.org/content/367/6479/791/suppl/DC1
Materials and Methods
Figs. S1 to S13
Tables S1 and S2
References (27–34)

27 May 2019; accepted 18 December 2019
10.1126/science.aay1790



Major role of particle fragmentation in regulating biological sequestration of CO₂ by the oceans

Nathan Briggs, Giorgio Dall'Olmo, and Hervé Claustre

Science **367** (6479), . DOI: 10.1126/science.aay1790

Breaking up is easy to do

Sinking particles transport carbon to the seafloor, where they are buried in sediments and either provide food for benthic organisms or sequester the carbon they contain. However, only #30% of the maximum flux reaches depths of a kilometer. This loss cannot be fully accounted for by current measurements. Briggs *et al.* used data collected by robotic Biogeochemical-Argo floats to quantify total mesopelagic fragmentation and found that this process accounts for roughly half of the observed flux loss (see the Perspective by Nayak and Twardowski). Fragmentation is thus perhaps the most important process controlling the remineralization of sinking organic carbon.

Science, this issue p. 791; see also p. 738

View the article online

<https://www.science.org/doi/10.1126/science.aay1790>

Permissions

<https://www.science.org/help/reprints-and-permissions>

Use of this article is subject to the [Terms of service](#)

Science (ISSN 1095-9203) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science* is a registered trademark of AAAS.

Copyright © 2020 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works